

⚡ ARDUINO SENSORS LAB PACK

5 Sensor Experiments | STEM Electronics & Coding
Grades 5–9 | Makerspace & STEM Classrooms

💡 Light Sensor

🌡️ Temperature Sensor

🔊 Ultrasonic Sensor

🔘 Push Button

📏 Potentiometer

ARDUINO SENSORS LAB PACK

5 SENSOR EXPERIMENTS | STEM ELECTRONICS & CODING
GRADES 5–9 | MAKERSPACE & STEM CLASSROOMS



LIGHT SENSOR



TEMPERATURE
SENSOR



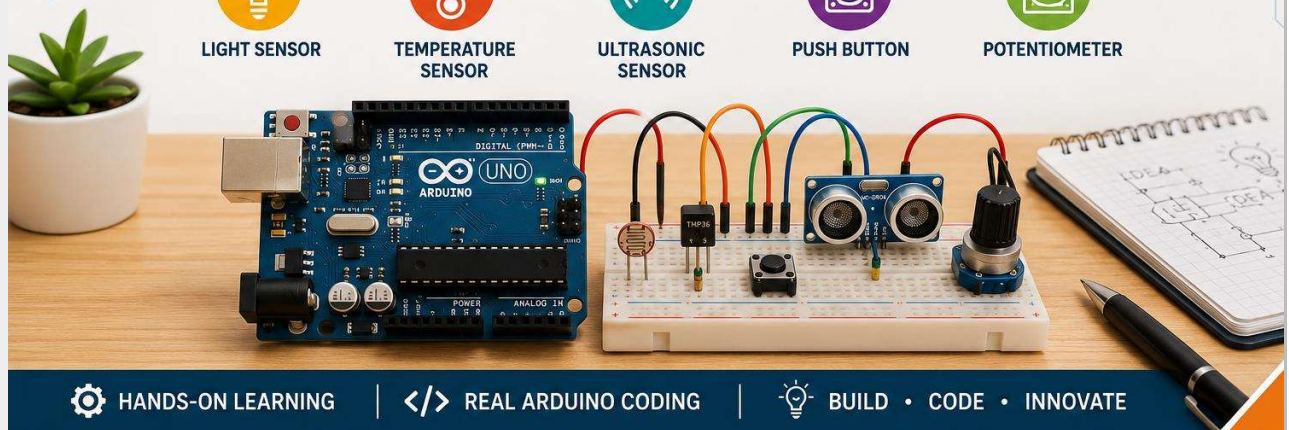
ULTRASONIC
SENSOR



PUSH BUTTON



POTENTIOMETER



HANDS-ON LEARNING



REAL ARDUINO CODING



BUILD • CODE • INNOVATE

What's Inside This Lab Pack?

- ✓ 5 Complete Sensor Lab Experiments with Circuit Guides & Arduino Code
- ✓ 5 Student Worksheets with Prediction & Observation Tables
- ✓ Teacher Quick Start Guide, Materials List & Safety Guidelines
- ✓ Standards Alignment (NGSS, CSTA, ISTE)
- ✓ Engineering Challenge: Build a Smart Device
- ✓ Assessment & Grading Rubric
- ✓ Teacher Answer Guide with Troubleshooting Tips
- ✓ Implementation Tips & Differentiation Suggestions

WHO THIS RESOURCE IS FOR

This lab pack is designed for educators and programs introducing students to Arduino, physical computing, and STEM electronics. It requires no prior Arduino experience from teacher or students.

Ideal for:

- STEM & Technology Teachers (Grades 5–9)
- Makerspace Facilitators
- Robotics Club Instructors
- Engineering Elective Teachers
- After-School STEM Programs
- Beginner Arduino Classrooms
- Summer Coding Camps

Key Details:

- Total Teaching Duration: 6–8 class periods (45–60 min each)
- Recommended Grade Levels: 5–9
- No prior Arduino experience required (teacher or student)
- Flexible: use all 5 labs or pick individual labs
- Works as stations, whole-class, or independent project
- Compatible with Tinkercad for no-hardware online simulation



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TEACHER QUICK START GUIDE

Read this page before distributing lab materials to students. Everything you need to prepare is listed here.

Required Materials Per Student Group (2–3 students)

Component	Quantity	Notes
Arduino Uno (or compatible)	×1	Any Uno-compatible board works
Half or full-size breadboard	×1	
Jumper wires (male-to-male)	20+	Assorted colors recommended
USB Type A to Type B cable	×1	Standard Arduino cable
Computer with Arduino IDE	×1	Free at arduino.cc/en/software
LED (any color)	×2	Red and green recommended
220Ω resistor	×2	For LED current limiting
10kΩ resistor	×1	For LDR and push button
LDR / Photoresistor	×1	Lab 1
TMP36 temperature sensor	×1	Lab 2 — handle orientation carefully
HC-SR04 ultrasonic sensor	×1	Lab 3
Push button (4-leg)	×1	Lab 4
10kΩ potentiometer	×1	Lab 5

⚠ Beginner Warnings & Common Setup Issues

⚠ Common First-Time Issues to Address Before Class

- Arduino IDE not recognizing the board: Check Tools > Board > Arduino Uno AND Tools > Port (select the COM port with Arduino listed)
- Upload fails silently: Make sure the Serial Monitor is CLOSED before uploading code
- TMP36 sensor gets hot immediately: Students installed it backwards — disconnect power and correct orientation before damage occurs
- LED does not light up: Most often a reversed polarity issue (longer leg = positive = connect to resistor, then pin)
- HC-SR04 returns 0 or wildly large numbers: TRIG and ECHO pin wiring is likely swapped in the circuit
- Breadboard connections feel loose: Wires must be inserted firmly — a slightly loose wire causes intermittent failures



BEGINNER WARNINGS

Common Setup Issues



Board not recognized?



Check **Board** and **Port**



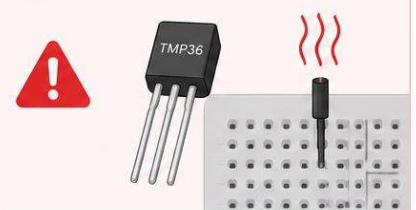
Upload fails?



Close **Serial Monitor**



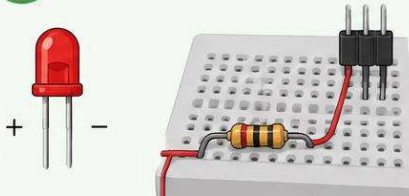
TMP36 gets hot?



Check **orientation**



LED not lighting?



Longer leg (+) to resistor, then pin



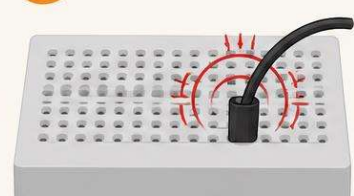
HC-SR04 readings wrong?



Check **TRIG** and **ECHO** wiring



Loose connections?



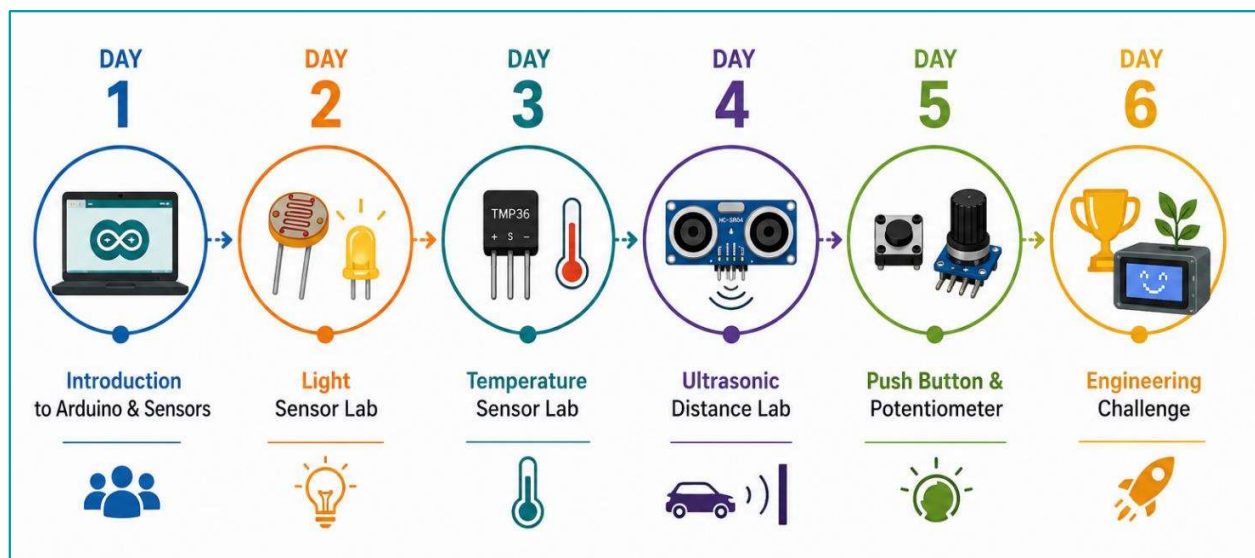
Insert wires **firmly**

Suggested 6-Day Lesson Flow

Day	Topic	Focus Activity
Day 1	Introduction to Arduino & Sensors	Arduino IDE setup, Blink example, class discussion on sensors in daily life
Day 2	💡 Light Sensor Lab	LDR circuit, automatic night light, Serial Monitor observation
Day 3	🌡️ Temperature Sensor Lab	TMP36 circuit, temperature monitor, math formula discussion
Day 4	🚗 Ultrasonic Distance Lab	HC-SR04 circuit, parking sensor, distance accuracy comparison
Day 5	🔘 Push Button & 📏 Potentiometer	Digital vs. analog input comparison, LED control, PWM dimming
Day 6	🏆 Engineering Challenge	Plan, build, and present a smart device using one chosen sensor

TEACHER TIP

On Day 1, spend 10 minutes showing students a completed circuit (e.g., a blinking LED) before explaining how it works. Curiosity-first learning boosts engagement significantly.



SECTION 5 — SENSOR LAB ACTIVITIES

Each lab includes a circuit diagram, circuit explanation, Arduino code, and step-by-step instructions. Students should complete the corresponding Student Worksheet (Section 6) alongside each lab.



TEACHER TIP

Print the lab pages and distribute one per group. Encourage students to read through the full lab before touching any components.



LAB 1 — LIGHT SENSOR LAB

Automatic Night Light — How Do Devices Know When It Gets Dark?

What This Sensor Does

A Light Dependent Resistor (LDR), also called a photoresistor, changes its electrical resistance based on how much light hits it.

In bright light, the resistance is low. In darkness, the resistance is high. The Arduino reads this as an analog value between 0 and 1023 using its analog input pins.

When the light level drops below a set threshold, the code switches an LED on — just like an automatic night light.

Real-World Examples

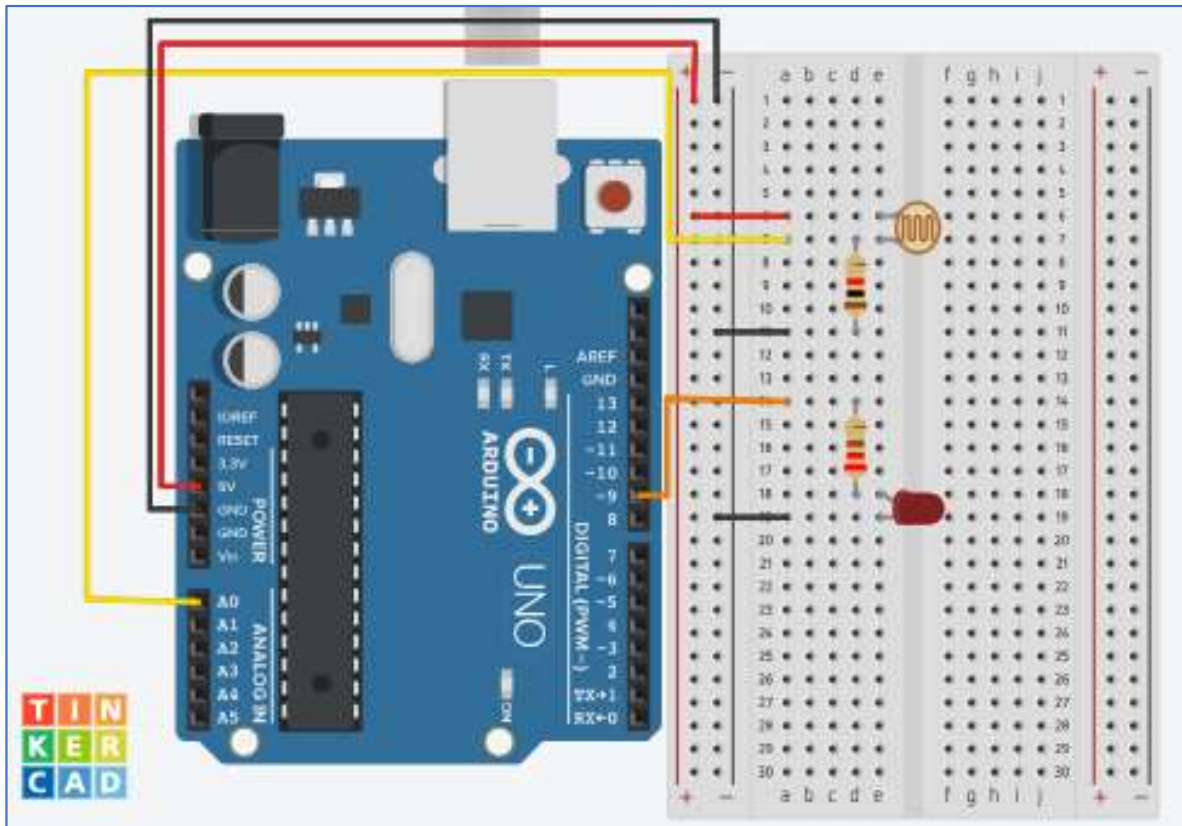
- Automatic street lights that turn on at dusk and off at sunrise
- Smartphones that adjust screen brightness based on room lighting
- Security cameras that switch to night-vision mode in darkness
- Solar garden lights that charge during the day and glow at night



Components Needed

- Arduino Uno (or compatible board) ×1
- Breadboard ×1
- Light Dependent Resistor (LDR / Photoresistor) ×1
- LED ×1
- 220Ω resistor ×1
- 10kΩ resistor ×1
- Jumper wires

Circuit Diagram



Circuit Explanation

The LDR and a 10kΩ resistor form a voltage divider circuit connected to analog pin A0.

In bright light: LDR resistance is low → voltage at A0 is HIGH → analog value is high (near 1023).

In darkness: LDR resistance is high → voltage at A0 is LOW → analog value is low (near 0).

An LED is connected to pin 9 through a 220Ω resistor. The code turns the LED on when the reading drops below a set threshold.



TEACHER TIP

The threshold value (400) is a starting point. Brighter classrooms may need a lower threshold. Have students experiment by changing this number — this is real engineering iteration!

Step-by-Step Instructions

1. Build the circuit as shown:
 - connect the LDR between 5V and A0
 - add a 10kΩ resistor from A0 to GND
 - connect the long leg of LED to pin 9 through a 220Ω resistor and short leg to GND.
2. Open the Arduino IDE, enter the code, and click **Verify** (✓) to check for errors.
3. Connect the Arduino board, then click **Upload** (→).
4. Open the Serial Monitor and set the baud rate to **9600**.
5. Cover the LDR with your hand and observe how the sensor values change.
6. Adjust the threshold value to control when the LED turns ON.
7. Test the circuit in different lighting conditions around the classroom.

Expected Sensor Readings — Lab 1

- Bright daylight / outdoors: ~900–1023
- Normal classroom lighting: ~500–800
- Dim room / desk lamp covered: ~200–450
- Fully covered with hand (dark): ~0–100
- LED should switch ON below your threshold — monitor Serial output to find the right value for your environment

Common Student Mistakes — Lab 1

- LED polarity reversed — the longer leg (positive/anode) must connect to the resistor side, not directly to GND
- Missing the 10kΩ pull-down resistor between A0 and GND — without it, readings will be random and unstable
- LDR connected only to 5V with no resistor — this creates a short circuit and gives a constant reading near 1023
- Threshold value too high or too low — encourage students to observe Serial Monitor values first, then set threshold in the middle
- `analogRead()` used on a digital pin number — must use A0, A1, A2... not 0, 1, 2

Arduino Code

Type this code into your Arduino IDE exactly as shown, then click Upload.

```
// LAB 1: Light Sensor - Automatic Night Light
int ldrPin = A0;      // LDR connected to analog pin A0
int ledPin = 9;       // LED connected to pin 9
int threshold = 400;  // Adjust this value for your environment

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int ldrValue = analogRead(ldrPin); // Read sensor
  Serial.print("Light level: ");
  Serial.println(ldrValue);

  if (ldrValue < threshold) {
    digitalWrite(ledPin, HIGH); // Dark: LED ON
  } else {
    digitalWrite(ledPin, LOW);  // Bright: LED OFF
  }
  delay(200);
}
```



Differentiation Options:

Easier Version Pre-wire the circuit for students who struggle with breadboards. Focus their effort on observing sensor values and answering reflection questions.	On-Level Complete the standard lab as written. Encourage students to explain what happens when they change the threshold.	Extension Challenge Add a second LED for a "dim warning" level, or use <code>analogWrite()</code> to make the LED fade in smoothly instead of switching abruptly.
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🚀 Challenge Extension

Build a **Smart Night Light** that automatically turns ON when it gets dark and OFF when it becomes bright.

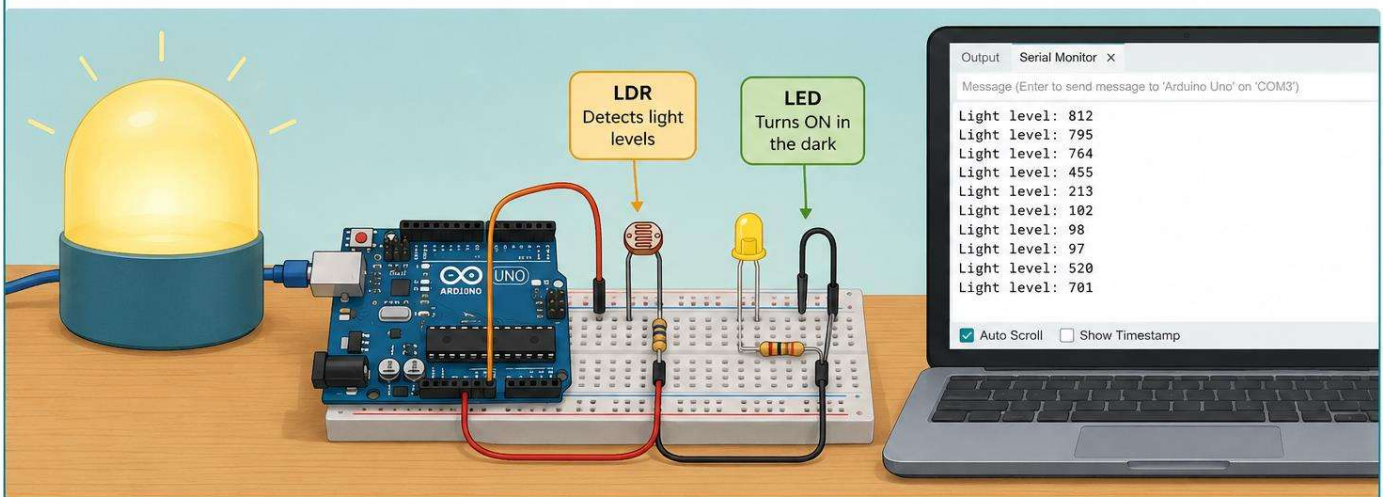
- Add a second LED with a different color
- Display "DARK" or "BRIGHT" in the Serial Monitor
- Add a buzzer that sounds when the room becomes dark
- Adjust the threshold value to change the light sensitivity

★ CHALLENGE EXTENSION

Build a "Smart Night Light" that turns on automatically when it gets dark and off when it's bright.

Ideas to enhance your project:

- Add a second LED with a different color.
- Use the Serial Monitor to show "DARK" or "BRIGHT".
- Add a buzzer to make a sound when it's dark.
- Change the threshold value to customize sensitivity.



Think Big! This same idea can be used in real-world devices like automatic street lights, security lights, and smart home systems.



Reflection Questions

1. At what sensor reading did the LED turn on for your group? Was it the same for other groups?

2. Why might different groups have different threshold values in the same classroom?

3. Name TWO devices in your school or home that use light sensors and explain how they work.

4. How would you change the code so the LED slowly fades in instead of switching on instantly?

QUIZ 1 – LIGHT SENSOR LAB

Name: _____ Date: _____

Part A — Multiple Choice

1. What does an LDR/photoresistor detect?
 - A. Sound
 - B. Light level
 - C. Temperature
 - D. Distance
2. What Arduino function reads values from A0?
 - A. digitalWrite()
 - B. analogRead()
 - C. delay()
 - D. pinMode()
3. In this lab, the LED turns on when the sensor reading is:
 - A. Below the threshold
 - B. Always 1023
 - C. Above 2000
 - D. Equal to the pin number
4. The LDR and 10kΩ resistor are used to make a:
 - A. Voltage divider
 - B. Motor driver
 - C. Speaker circuit
 - D. Battery charger
5. Why might two groups need different threshold values?
 - A. Different classroom light levels
 - B. The Arduino has no memory
 - C. LEDs do not need resistors
 - D. Serial Monitor changes wire colors

Part B — Short Answer

6. Explain what happens to the sensor reading when you cover the LDR with your hand.

7. Name one real-world device that could use a light sensor and explain why.



SECTION 7 — ENGINEERING CHALLENGE

Build a Smart Device Using a Sensor

Apply your knowledge from all 5 labs to design and build your own sensor-powered invention!

The Challenge

You have explored 5 different types of sensors. Now it's your turn to be the engineer! Choose ONE sensor from the labs and design a useful real-world device that solves a problem or improves someone's life.

Your device must:

- Use at least one sensor (from any of the 5 labs)
- Include at least one output (LED, buzzer, or Serial Monitor message)
- Solve a real problem or perform a useful function
- Have working Arduino code that makes the sensor and output interact

Step 1: Choose Your Sensor & Idea

Circle the sensor you will use:



Light Sensor



Temperature



Ultrasonic



Push Button



Potentiometer

My device idea is:

ASSESSMENT & GRADING RUBRIC — Engineering Challenge

FOR TEACHERS

Use this rubric to assess the Engineering Challenge (Section 7). Share it with students before the challenge begins so they understand the expectations. Adjust point values as needed for your grading system.

Engineering Challenge Rubric

Category	4 – Excellent	3 – Proficient	2 – Developing	1 – Beginning
Circuit Construction	All connections correct; clean, stable wiring; correct resistors used.	Minor wiring errors fixed quickly; circuit functional.	Multiple errors; required teacher assistance to complete circuit.	Circuit incomplete or non-functional.
Code Functionality	Code compiles and runs correctly; sensor data displayed accurately.	Code runs with minor debugging; mostly correct output.	Code has errors requiring help; partial functionality.	Code does not compile or produce any output.
Creativity	Unique and inventive device idea with clear real-world application.	Solid idea with some originality.	Basic idea, closely follows the example given.	No clear concept or copied directly from example.
Problem Solving	Independently identified and resolved all issues.	Resolved most issues with minimal hints.	Needed regular support to troubleshoot problems.	Unable to troubleshoot without direct teacher guidance.
Documentation	Worksheet fully complete; detailed, thoughtful answers.	Worksheet mostly complete; adequate explanations.	Worksheet partially complete; answers brief or unclear.	Worksheet incomplete or missing.
Teamwork	All members contributed equally; excellent communication.	Most members contributed; collaboration was positive.	Some members dominated; limited collaboration.	Little to no evidence of teamwork.

ABOUT THE AUTHOR

Founder of Math-O-Tronics

Ahmad Abdullah

Ahmad Abdullah is a STEM educator with a passion for combining mathematics, technology, and hands-on engineering to create meaningful learning experiences for students.



Bachelor of Science in Mathematics / Education — Lebanese International University



Certified Robotics Trainer — Global Board Lebanon

Professional Background & STEM Skills

Ahmad has a strong academic background in Mathematics and Technology. In addition to his formal education, he continuously develops his skills in:

- electronics
- programming
- robotics
- Arduino systems
- educational technology

He believes in lifelong learning and regularly explores new teaching tools, STEM strategies, and project-based learning methods to help students develop confidence, curiosity, and critical thinking skills.

Ahmad is especially passionate about helping students understand mathematics through clear, logical instruction connected to real-world applications.

STEM Activities & Interests

- building electronics projects
- coding with Arduino
- designing robotics activities
- creating classroom-ready STEM resources

His mission through Math-O-Tronics is to provide practical, classroom-tested materials that make teaching easier and learning more engaging for students.